

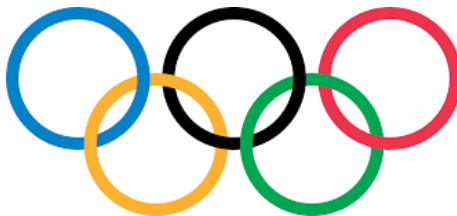
Analysis of Olympic Medal Counts

Svetomir Marić 81m/23
dmi.81m.23@student.pmf.uns.ac.rs

Faculty of Sciences, University of Novi Sad

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About Olympic Games



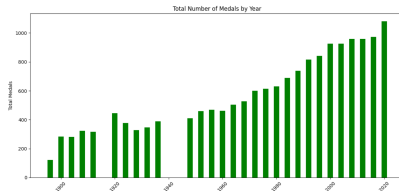
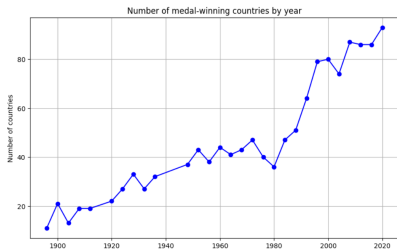
The Olympic Games represent the biggest sporting event, bringing together the most diverse range of sports in one city, all within two weeks.

Olympic Medals



- From the early years until today, a large number of medals have been awarded. The countries winning the most medals have changed over time.
- Many factors have influenced the distribution of medals over the years (expensive travel in the early years, wars, politics, etc.).
- The main goal of my project is to analyze the number of medals and to examine the impact of hosting the Olympic Games on the number of medals won.

Number of medals and winners through the years



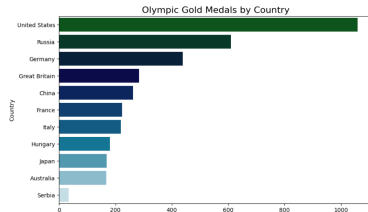
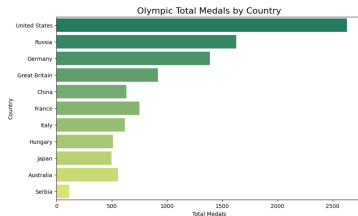
- The number of medals awarded at the Olympics has changed over the years, generally showing an upward trend.
- Similarly, the number of medal-winning countries has also increased, with more and more nations winning medals.

Median



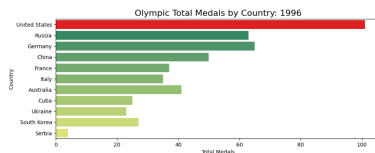
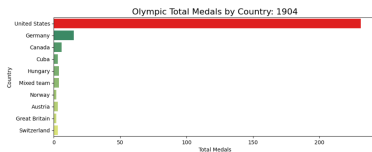
- Although the number of participants winning medals has increased, it is interesting to note that the median(of number of medal winners) has not changed much.
- Apart from a few peaks, the median is generally within the range of 3 to 6.

The best of all time



- The graphs display the greatest performers of all time and Serbia.
- The ranking is based on the number of gold medals won.
- The ranking would be different if the total number of medals were considered.

USA 1904 VS USA 1996



- In the early years, hosting was crucial; hosts always took the largest share of medals in that year.
- Today, however, the advantage of hosting is not as pronounced.
- Here, we can see how this was the case for the USA in 1904 and 1996.

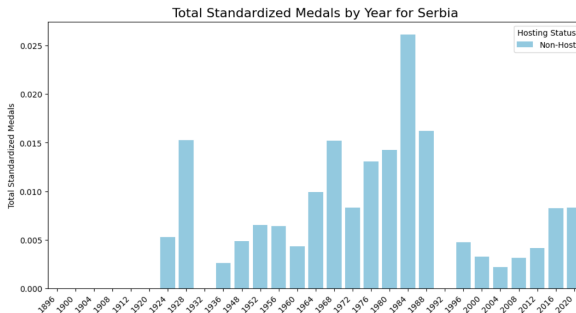


Figure: Distribution of medals for Serbia

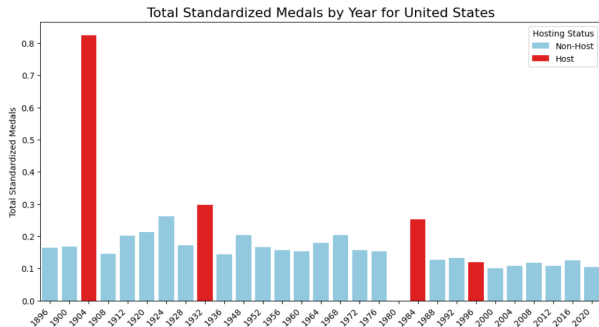


Figure: Distribution of medals for USA

- Here, we see how the USA creates local peaks in the years of hosting.

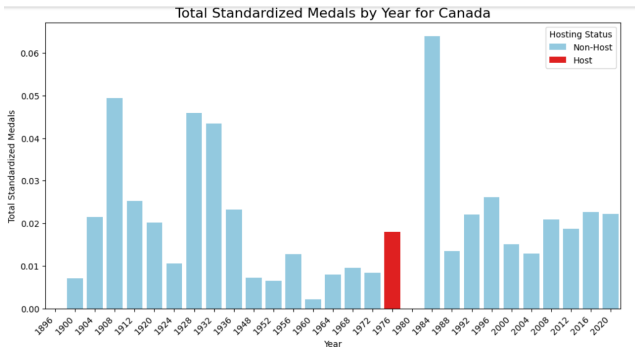
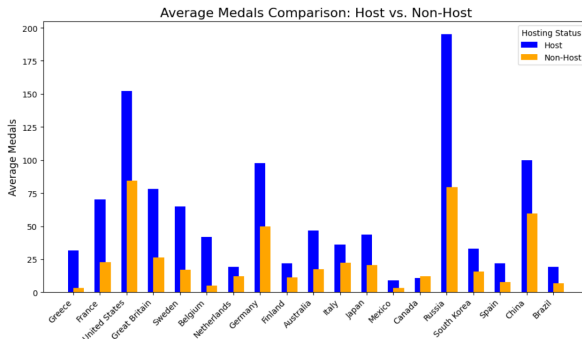


Figure: Distribution of medals for Canada

- However, there is also a completely opposite example. Canada recorded one of its worst results in the year it hosted the Games.

Main Goal - Host Advantage



- Here, we can see the average for countries when they are hosts compared to years when they are not.
- All data are standardized when comparing different years.

- The dataset includes the number of medals (gold, silver, bronze) that countries have won at the Olympic Games from 1896 to 2020.
- This data set is provided in the CSV format at Kaggle.com.
- Here is the link
<https://www.kaggle.com/datasets/ramontanoeiro/summer-olympic-medals-1986-2020/data>